

Learning for Inflation Forecasting with Dynamic Feature Spaces

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Abstract—Inflation forecasting becomes especially challenging during structural breaks, such as the COVID-19 pandemic, when historical patterns are disrupted. We evaluate lightweight ML strategies for adapting to dynamic feature spaces, including rolling-window training, interpolation, and reversible normalization. Using U.S. CPI data and pandemic-specific features (cases, deaths, vaccines, hospitalizations), we compare traditional econometric models, tree ensembles, and deep sequential architectures. We find that ensemble methods deliver the most accurate forecasts (RMSE 0.36 with interpolated training data, 0.61 without), while normalization layers improve neural networks’ robustness under distribution shifts. These results show that simple, low-cost adaptation methods sustain forecast reliability during shocks, aiding policymakers.

Index Terms—COVID-19, Dynamic Feature Expansion, Inflation Prediction, Temporal Prediction

I. INTRODUCTION

The COVID-19 pandemic of 2020 had profound effects on the global economy [7]. The resulting lockdowns caused severe contractions in the restaurant, tourism, and travel industries, while stock markets experienced dramatic declines—falling by 37% in the Dow Jones Industrial Average and 34% in the S&P 500 during March 2020 [16]. The pandemic was followed by a sharp rise in inflation, with the Consumer Price Index (CPI) recording its fastest increase in over three decades [10].

Accurate inflation forecasts are vital for monetary policy during crises. Reliable forecasts enable central banks to anticipate inflationary shocks and implement timely interventions, mitigating adverse effects on lower-income households most vulnerable to rising prices. Yet forecasting inflation during structural breaks, when relationships among variables shift, remains difficult.

Machine learning (ML) models have demonstrated strong performance in inflation prediction, often surpassing traditional econometric methods [2], [18]. Yet, most prior work assumes stable data distributions and a fixed set of explanatory variables. During extreme events such as the COVID-19 pandemic, new predictors (e.g., case counts, deaths, vaccinations) emerge suddenly, creating a *dynamic feature space*. Existing forecasting models are not designed to incorporate these evolving features without extensive retraining, leading to degraded performance and limited adaptability. This highlights a critical

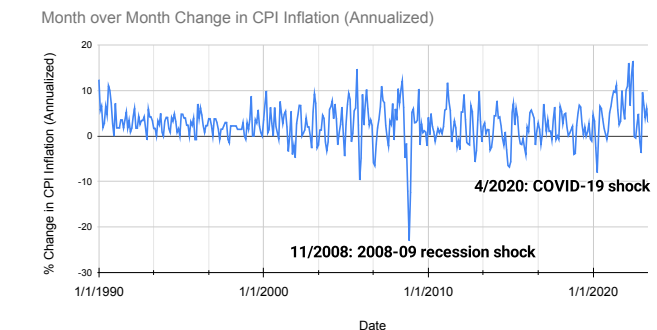


Fig. 1. Annualized month-over-month percent change in CPI inflation from January 1990 to May 2023.

research gap: how can ML models remain robust and adaptable when both the data distribution and feature space evolve during structural breaks?

We address this gap by evaluating efficient ML strategies for inflation forecasting under dynamic features from the COVID-19 shock. The main objectives of this work are to:

- Assess the performance of diverse ML models in predicting CPI inflation changes before, during, and after the pandemic to identify which approaches best handle abrupt structural breaks.
- Enhance predictive accuracy by integrating pandemic-specific variables (e.g., deaths, vaccine doses) with traditional macroeconomic indicators.
- Develop and test lightweight adaptation strategies - including rolling-window training, data interpolation, and normalization layers - to mitigate challenges from limited data and shifting feature distributions.
- Offer insights for future research on adaptive forecasting frameworks that maintain robustness under dynamic feature spaces.

By systematically examining these strategies, this work contributes to developing resilient ML-based forecasting methods capable of sustaining accuracy when confronted with shocks that alter both data distributions and input feature structures.

II. RELATED WORK

Research on inflation forecasting spans both classical econometric approaches and modern machine learning (ML) or hybrid methods. Traditional time-series and structural models—such as ARIMA, VAR, factor models, and time-varying parameter regressions—remain foundational (e.g. [13], [23]). At the same time, ML and data-rich approaches have gained traction as computational tools and datasets have expanded (e.g. [5], [21]). While these studies demonstrate the growing strength of ML forecasting, most still assume fixed feature spaces and do not explicitly address how models should adapt when new variables emerge during structural breaks.

A. Inflation Forecasting During COVID-19

Bobeica and Hartwig [4] showed that COVID-19 disrupted VAR forecasts in Europe and that incorporating external information, such as professional inflation expectations, improved forecast performance. Yuniarti et al. [22] found a negative relationship between COVID-19 cases and inflation in Indonesia, highlighting the importance of adapting models during extreme shocks. These results illustrate how exogenous, time-varying variables become newly relevant during crises, posing challenges for models trained on static feature sets.

B. Machine Learning Approaches to Inflation

Machine learning techniques have increasingly been applied to inflation forecasting, often outperforming traditional econometric models. For instance, Aras and Lisboa [1] use tree-based ensemble models with Shapley-value explanations to improve transparency in inflation forecasting. Araujo and Gaglianone [2] benchmark a wide range of ML models—including random forests, boosting, and neural networks—against classical approaches for Brazilian inflation. Medeiros [14] develops data-rich ML forecasting models combining shrinkage and nonlinear methods for macroeconomic prediction. Kohlscheen [13] applies regression-tree ensembles across advanced economies to identify inflation drivers in a data-driven framework. Paranhos [15] employs recurrent neural networks (LSTM) to forecast inflation and extract latent temporal features, while Vargas [17] evaluates KNN, random forest, boosting, and LSTM models for Costa Rican inflation. Singh [18] demonstrates the effectiveness of tree-based algorithms for capturing nonlinearities in inflation dynamics. Finally, Theoharidis et al. [19] propose hybrid deep-learning architectures (variational autoencoders and ConvLSTM) to capture complex temporal dependencies in inflation forecasting. Despite their diversity, these studies share a limitation: nearly all assume a fixed feature space and lack mechanisms to integrate newly emerging predictors during structural breaks.

C. Dynamic Features and Distribution Shifts

Beyond inflation-specific work, the broader time-series and ML literature explores ways to model evolving input spaces and handle distributional changes. Examples include reversible normalization techniques [12], adaptive regularization and dynamic variable selection [3], and online or rolling-update

algorithms. While such adaptive methods are well-established in other domains, they have rarely been applied to inflation forecasting under structural breaks. This gap motivates our focus on simple, general-purpose strategies for dynamic feature expansion in economic settings.

D. Summary

Prior research shows that ML and hybrid methods can enhance forecast accuracy relative to classical econometric models but often struggle under structural breaks. Econometric models typically handle shocks through explicit structural assumptions, whereas ML models emphasize predictive flexibility but assume static input sets. Few studies have explored how to update models when entirely new variables emerge—such as COVID-19 cases, deaths, and vaccinations. Our work addresses this gap by systematically testing lightweight adaptation strategies for dynamic feature spaces, showing how models can incorporate new information streams without retraining from scratch.

III. METHODS

A. Problem Formulation

Assuming that we are given a historical time-series feature matrix $X_{1:t-1}$, the goal of this study is to predict the change of inflation rate from time $t-1$ to time $t+d$ where d is the leadtime selected from $0, 1, \dots, D$, which can be expressed in a general form as:

$$\Delta_{t+d} = f(X_{1:t-1}) + \epsilon, \quad (1)$$

where Δ_t is the change in the inflation rate from time $t-1$ to time $t+d$, $f(\cdot)$ is a function representing the model's prediction utilizing the economic and COVID-19 data as input, and ϵ is the error in the prediction. $X_{1:t-1}$ is a matrix of the economic and COVID-19 data up to time $t-1$, consisting of time-series data of economy-related measurements and COVID-19 features. For each time stamp, X_i is a collection of multiple elements $X_i = (e_1, e_2, \dots, e_n, c_1, c_2, \dots, c_m)$, where each entry e_i is a macroeconomic-related feature and each entry c_i is a COVID-related feature. In the real-world setting, each time stamp corresponds to a month by nature of the data we use, as we describe in the following section.

B. Data Sources and Features

To train our models, we use a baseline set of economic features. Our feature selection was guided by prior literature, domain knowledge of inflation drivers, and an empirical assessment of inter-variable relationships. We examined correlation matrices to identify redundant features and retain variables that provided unique predictive signals. The final set includes food and energy prices, housing and labor market indicators, and consumer expectations, all of which are historically tied to inflation dynamics and available at consistent monthly frequency. To capture the structural break during COVID-19, we also incorporated pandemic-specific variables such as cases, deaths, hospital admissions, and vaccine doses. The final set of features includes: *Chicken Price*, *Electricity*

Price, Gasoline Price, Utilities Price, Housing Starts, Housing Price Index, Rental Price Average, Industrial Production Total Index, Unemployment Rate, University of Michigan Inflation Expectations Survey, COVID Deaths, COVID Vaccine Doses Administered, COVID Hospital Admissions, COVID Cases, and Consumer Price Index (CPI) [6], [8], [9], [20].

Incorporating COVID-19 data is crucial for maintaining model accuracy during the pandemic. As shown in Figure 1, the change in inflation generally follows a roughly sinusoidal pattern, but two major inflation shocks can be identified: one during the 2008-09 recession (the first annotation) and the second during COVID-19 (the second annotation). This second shock began with the pandemic in 2020 and continued into 2021 and 2022. We deduce that the models can leverage COVID-19 data to comprehend the relationship between the rise of the pandemic and the resulting inflation shock, thereby enhancing model accuracy during this period. However, due to the unavailability of COVID-19 data pre-pandemic, we need to build models that can adapt to a dynamic feature space that varies over time. We address this challenge with two approaches. In the first approach, we set the COVID-19 data values to 0 for the period before the pandemic, allowing us to test which models are most accurate in predicting changes in inflation. Alternatively, for some models, we implement a new adaptive training approach to handle the dynamic feature space, as detailed in Section III-C.

For our data preprocessing, we calculate the month-over-month percent change in the data and then standardize it to remove the mean and scale the data to unit variance. To determine the appropriate number of months to lag our features, we utilize partial autocorrelation, which helps detect any seasonal or cyclic trends that can be leveraged. Additionally, we create an alternative version of the dataset using linear interpolation to convert the data from a monthly to a weekly frequency, effectively increasing the amount of data by a factor of four. We refer to this version as the interpolated dataset.

C. Adaptive Training with Dynamic Features

In real-world forecasting scenarios, new explanatory variables may emerge over time—for example, COVID-19 case counts and vaccination data became available only after 2020. This results in a *dynamic feature space*, where the dimensionality of the input feature vector changes as new information sources appear. Conventional models must typically be retrained from scratch to incorporate such new features, which is computationally expensive and impractical for time-sensitive applications.

To address this limitation, we propose an *adaptive training approach* that allows models to expand their feature space without full retraining. The key idea is to preserve the parameters associated with previously learned features and initialize new parameters only for the added variables, maintaining model continuity while allowing structural adaptation.

For example, consider an OLS regression model:

$$y = \alpha + \mathbf{x}^\top \boldsymbol{\beta} + \epsilon, \quad (2)$$

where y is the target variable (e.g., the change in CPI inflation), α is the intercept, $\mathbf{x} = [x_1, x_2, \dots, x_k]^\top$ is the feature vector with k elements, $\boldsymbol{\beta} = [\beta_1, \beta_2, \dots, \beta_k]^\top$ is the vector of regression coefficients, and ϵ is the residual error term.

When the feature space expands with n new variables—e.g., pandemic-related indicators—the model parameters must also expand. We redefine the feature and coefficient vectors as:

$$\mathbf{x}' = [x_1, x_2, \dots, x_k, x_{k+1}, \dots, x_{k+n}]^\top, \quad (3)$$

$$\boldsymbol{\beta}' = [\beta_1, \beta_2, \dots, \beta_k, \beta_{k+1}, \dots, \beta_{k+n}]^\top. \quad (4)$$

The original coefficients $[\beta_1, \dots, \beta_k]$ are preserved from the previously trained model, while the new coefficients $[\beta_{k+1}, \dots, \beta_{k+n}]$ are initialized randomly or using small prior values. The updated model becomes:

$$y' = \alpha + \mathbf{x}'^\top \boldsymbol{\beta}' + \epsilon', \quad (5)$$

where training resumes, allowing the model to adapt to the expanded feature space without discarding prior knowledge.

This approach can be generalized to nonlinear models such as feedforward neural networks by expanding the input layer to accommodate the new features. The existing weight matrices and biases are retained, and additional weights connecting the new inputs are initialized randomly:

$$\mathbf{W}' = [\mathbf{W} \quad \mathbf{w}_{\text{new}}], \quad (6)$$

where $\mathbf{W}$ represents the pretrained weights corresponding to the original inputs, and $\mathbf{w}_{\text{new}}$ contains the randomly initialized weights for the new features. This lets the network learn new relationships without disturbing prior ones.

We evaluate this adaptive training strategy against a baseline approach where pandemic-related variables are held constant (set to zero) in the pre-COVID period. The results of these methods are compared in Section V-D.

D. Single-temporal Point Modeling

In real-world scenarios, it can be challenging to acquire sufficient data for a new feature needed by a model, or there may be limited data available at the time for a specific variable. For example, with the onset of COVID-19, there was little COVID-19-related data available for training models. Therefore, we aim to assess the performance of single-timestep models in predicting changes in inflation. If a model can accurately predict changes in inflation based on a single timestep of data, it suggests that in future pandemic-like events, new variables arising from such events can be quickly incorporated into models to enhance their performance.

With this in mind, we refine our general problem formulation in Equation 1 to:

$$\Delta_t = f(X_{t-1}) + \epsilon,$$

where the model solely utilizes data from the previous timestep ($t - 1$) to predict the change in inflation from time $t - 1$ to time t .

The single-temporal point models we test include linear regression (LR), random forests (RF), and feedforward neural

networks (NN). Additionally, we develop modified versions of the LR and NN models to test the adaptive training approach we devised, referred to as DynLR and DynNN. We also evaluate the Reversible Instance Normalization (RevIN) neural network layers [12] in a model called NN-RevIn. This model includes an additional normalization layer for the input and a denormalization layer for the output, both symmetrically structured. These layers help remove non-stationary information, improving the model’s adaptability to distribution shifts, such as shocks in time series data. Therefore, we test its performance in adapting to the shocks caused by the COVID-19 pandemic.

E. Multi-temporal Point Modeling

While single-temporal point models can be beneficial in scenarios where limited data is available for necessary features, multi-temporal point models can capture relationships over time between variables in time series data more effectively and often outperform single-timestep models. Therefore, we evaluate the performance of several common multi-temporal models. To do this, we refine our general problem formulation in Equation 1 to:

$$\Delta_t = f(X_{t-n-1:t-1}) + \epsilon$$

In this formulation, the model uses data from timesteps $t-n-1$ to $t-1$, where n is the length of the input sequence, to predict the change in inflation from $t-1$ to t .

The multi-temporal point models we test are vanilla Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM) networks, and Gated Recurrent Unit (GRU) networks.

F. Ensemble Models

Ensemble learning combines multiple base models to generate a single, more reliable prediction by leveraging the diversity of individual learners. The central idea is that aggregating outputs from heterogeneous models can reduce variance and improve generalization, yielding performance that surpasses any individual component model.

In this study, the ensemble consists of three lightweight base regressors: Linear Regression (LR), Random Forest (RF), and Feedforward Neural Network (NN). These models were selected because they are computationally efficient, conceptually simple, and do not rely on multi-timestep sequences, allowing us to directly compare their combined performance against more complex temporal models such as RNN, LSTM, and GRU.

Each base model is trained independently on identical training and validation splits using the same feature pre-processing and normalization procedures. No information is shared between models during training. Let $f_i(\mathbf{x})$ denote the prediction of the i -th model for an input feature vector $\mathbf{x}$. The ensemble’s output is then computed using an unweighted averaging scheme:

$$\hat{y}_{\text{ens}} = \frac{1}{M} \sum_{i=1}^M f_i(\mathbf{x}), \quad (7)$$

where $M = 3$ is the number of constituent models. This approach assigns equal weight to each model’s prediction, ensuring that no single model dominates the final output.

The ensemble was implemented in a fully parallel manner: each base learner generates predictions independently for every test window, after which the predictions are averaged element-wise to obtain the ensemble forecast. All evaluations are performed under the same rolling-window framework used for individual models.

IV. EXPERIMENTAL SETUP

In our study, we utilize an online learning approach to train the models over time using rolling windows. This approach allows us to evaluate how well our models perform before COVID-19 and how their performance changes when the COVID-19 structural break occurs. Our setup for the rolling windows varies between tests conducted on the original dataset and the interpolated dataset. The interpolated version, having more training samples, enables the use of more windows with larger sample sizes for training the model. To ensure consistency between the setups for the interpolated and non-interpolated datasets, we group the timeline into three main sections for both datasets: a “pre-2020” section (before COVID-19); a small, single-window “transition” section to let the model initially observe the very beginning of COVID-19’s appearance at the end of 2019/start of 2020; and finally, a “post-2020” section, which includes the true onset of the COVID-19 pandemic and all subsequent data.

To counteract the small sample sizes in each window when training the models, we weigh the training samples based on their temporal recency. This approach ensures that as the model trains, it places greater emphasis on more recent samples. As the model moves from window to window, it should better capture the trend of inflation changes over time. The sample weight for any given sample w_i is calculated as: $w_i = i \times \frac{c}{\sum_{j=1}^n j}$, where i is the index of the sample in the training set of n samples, and c is a constant. This method assigns greater weight to samples later in time (i.e., those with larger indices), enhancing their influence during training.

For each model, during each testing window, we calculate the root mean squared error (RMSE) and mean absolute error (MAE) using their traditional formulas, as defined in [11].

To aggregate the metrics for each model, we first compute the average of each metric within the time sections (e.g., in the post-2020 period). Next, to obtain the average performance of each model across all periods, we use a weighted average of the average performance in each period, with weights based on the testing window size of each period:

$$\frac{n_{\text{pre}} \cdot \mu_{\text{pre}} + n_{\text{trans}} \cdot \mu_{\text{trans}} + n_{\text{post}} \cdot \mu_{\text{post}}}{n_{\text{pre}} + n_{\text{trans}} + n_{\text{post}}},$$

where each μ and n correspond to the mean and sample size of that testing period, respectively.

When working with the **interpolated weekly dataset**, we only calculate evaluation metrics using the original, observed monthly data points (all synthetic samples generated via linear

TABLE I
COMPARISON OF INFLATION CHANGE PREDICTION RESULTS;
HORIZONTAL LINE SEPARATES SINGLE TEMPORAL POINT METHODS FROM
MULTI-TEMPORAL POINT METHODS

Model	Non-interpolated		Interpolated	
	Test RMSE	Test MAE	Test RMSE	Test MAE
LR	0.9	0.92	0.59	0.59
DynLR	0.88	0.75	1.37	1.24
RF	0.77 (0.01)	0.78 (0.01)	0.69 (0.02)	0.66 (0.02)
NN	0.85 (0.04)	0.82 (0.03)	0.76 (0.05)	0.73 (0.05)
DynNN	1.03 (0.19)	1.01 (0.21)	0.84 (0.06)	0.77 (0.04)
NN-RevIn	0.87 (0.07)	0.74 (0.05)	0.76 (0.06)	0.71 (0.06)
Ensemble	0.61 (0.02)	0.63 (0.01)	0.36 (0.03)	0.33 (0.03)
RNN	0.86 (0.04)	0.79 (0.03)	0.8 (0.13)	0.77 (0.12)
LSTM	0.85 (0.07)	0.84 (0.07)	0.69 (0.08)	0.63 (0.07)
GRU	0.82 (0.06)	0.81 (0.03)	0.80 (0.16)	0.75 (0.16)

interpolation are *excluded* from RMSE and MAE calculations). In other words, the interpolated points serve only as additional model input to increase the effective sample frequency, but **do not contribute to the error computation**.

For multi-temporal point models (RNN, LSTM, GRU), we compared different timestep sequence lengths and selected 12-sample sequences for the regular experiments, based on optimal performance. Each model was run three times, and we report the mean RMSE and MAE along with the standard deviation.

Finally, we compare the complexity of neural-network-based models across both datasets, as determined after tuning their hyperparameters. Model complexity is crucial for comparing performance. The NN-RevIN is by far the most complex model, with 12,513 parameters, significantly more than the next largest models, the LSTM with 3,802 parameters and the RNN with 2,925 parameters. The regular NN and the GRU have the smallest number of parameters among the network-based models, with 2,085 and 2,160 parameters, respectively.

V. RESULTS

A. Without Data Interpolation

When trained on the non-interpolated dataset, most models predicted changes in inflation with relatively small error, though notable differences in performance emerged. Ranked by RMSE, the ensemble model achieved the best performance (0.61), followed by RF (0.77), GRU (0.82), and LSTM (0.85). In terms of MAE, the NN-RevIN model performed strongly (0.74), second only to the ensemble (0.63). While LR struggled (RMSE 0.90, MAE 0.92), combining models or adding normalization layers improved robustness even without interpolation.

The gap between ensemble and single models highlights the challenges of forecasting under structural breaks. Simple models tend to underfit, while neural networks risk overfitting given limited data. By pooling predictions, the ensemble balanced these trade-offs, yielding the most stable and accurate forecasts across time periods.

B. With Data Interpolation

When trained on the interpolated dataset, all models improved, though the degree of improvement varied. The ensemble once again achieved the strongest performance (RMSE 0.36, MAE 0.33), benefiting from the additional synthetic samples. Linear regression improved substantially, with RMSE dropping from 0.90 to 0.59, making it the second-best individual model. This suggests that linear interpolation strengthened the linear relationships LR is best suited to capture. RF also improved, with RMSE decreasing from 0.77 to 0.69.

Among neural architectures, LSTM achieved an RMSE of 0.69, outperforming RNN (0.80) and GRU (0.80). NN-RevIN maintained strong performance (RMSE 0.76, MAE 0.71), consistently surpassing the plain NN. These results indicate that interpolation benefits simpler and linear models the most, while neural models already capable of capturing nonlinear dependencies gained smaller but still meaningful improvements.

C. Overfitting of Neural Network Models

Though recurrent architectures should capture temporal dependencies, RNN, LSTM, and GRU often performed no better than the feedforward NN. For example, on the non-interpolated dataset, RNN (RMSE 0.86) and LSTM (0.85) were comparable to the plain NN (0.85), while GRU only marginally outperformed (0.82). On interpolated data, RNN and GRU performed worse (0.80) than LSTM (0.69). This likely reflects overfitting: recurrent networks have more parameters than LR, RF, or NN, but the dataset remained small.

Interestingly, NN-RevIN avoided this pitfall despite its higher parameter count (12,513). The normalization layers helped stabilize training and reduce sensitivity to distribution shifts, allowing the model to outperform the plain NN on both datasets. This suggests that architectural modifications aimed at mitigating non-stationarity may be more effective than simply increasing temporal modeling capacity when data are limited.

D. Adaptive Training with Dynamic Features

Adaptive training produced mixed results. For linear regression, expanding the coefficient vector when new features appeared yielded measurable improvements on the non-interpolated dataset: RMSE decreased from 0.90 to 0.88, and MAE from 0.92 to 0.75. These gains demonstrate that even simple adaptive strategies can help models leverage new information without retraining from scratch. However, DynLR performed poorly on the interpolated dataset (RMSE 1.37), suggesting that interpolation may have disrupted the balance between pre- and post-COVID features.

For neural networks, adaptive training was less successful. DynNN consistently underperformed the baseline NN across both datasets and exhibited the largest variance across runs (RMSE 1.03 ± 0.19 on non-interpolated data). This instability indicates that neural models are highly sensitive to initialization and hyperparameters when features expand dynamically. More sophisticated adaptive strategies, such as progressive

neural architecture growth or transfer learning approaches, may be needed to stabilize performance in these settings.

E. Comparative Analysis

Model performance reflected both feature-space characteristics and model complexity. Interpolation yielded the greatest gains for simpler models - LR improved from RMSE 0.90 to 0.59 and RF from 0.77 to 0.69 - by reinforcing low-order relationships in smoothed data. Neural models benefited less, though NN-RevIN and LSTM still improved modestly.

Recurrent architectures (RNN, LSTM, GRU) rarely outperformed the single-step NN and often overfit due to higher parameter counts. LSTM's interpolated RMSE of 0.69 was strong, but RNN and GRU lagged. NN-RevIN, despite its larger size, remained competitive by mitigating distribution shifts, underscoring the robustness benefits of normalization layers.

Adaptive training produced mixed results: DynLR slightly improved on LR in the non-interpolated data (RMSE 0.88 vs. 0.90) but degraded after interpolation, while DynNN remained unstable. These outcomes suggest that neural models are more sensitive to initialization and tuning under dynamic features.

The ensemble consistently delivered the best forecasts - RMSE 0.61 (non-interpolated) and 0.36 (interpolated) - by combining the stability of simple models with neural flexibility. In sum, interpolation mainly aids simpler models, recurrent networks risk overfitting with limited data, and ensembles and normalization layers provide the strongest resilience to structural breaks.

VI. CONCLUSION

This study presents practical solutions for handling dynamic feature spaces in time series forecasting, using COVID-19-era inflation as a case study. Combining traditional and pandemic-specific indicators, we compared multiple machine-learning models under structural breaks. Ensembles delivered the most accurate and stable forecasts, with RF, NN-RevIN, and LSTM as strong alternatives.

Lightweight strategies—interpolation, rolling windows, adaptive training, and normalization—kept models robust during shocks while avoiding costly retraining. Such methods can aid policymakers by supporting reliable forecasts during crises.

These insights extend to domains with rapidly evolving variables, such as climate, health, or finance. Evaluating approaches for feature expansion and distribution shifts offers a template for adaptive ML under uncertainty. Future work should pursue real-time streaming updates and advanced ensemble methods (e.g., Dynamic Ensemble Selection) to further enhance resilience. Robust forecasting ultimately requires both accurate models and adaptable training.

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